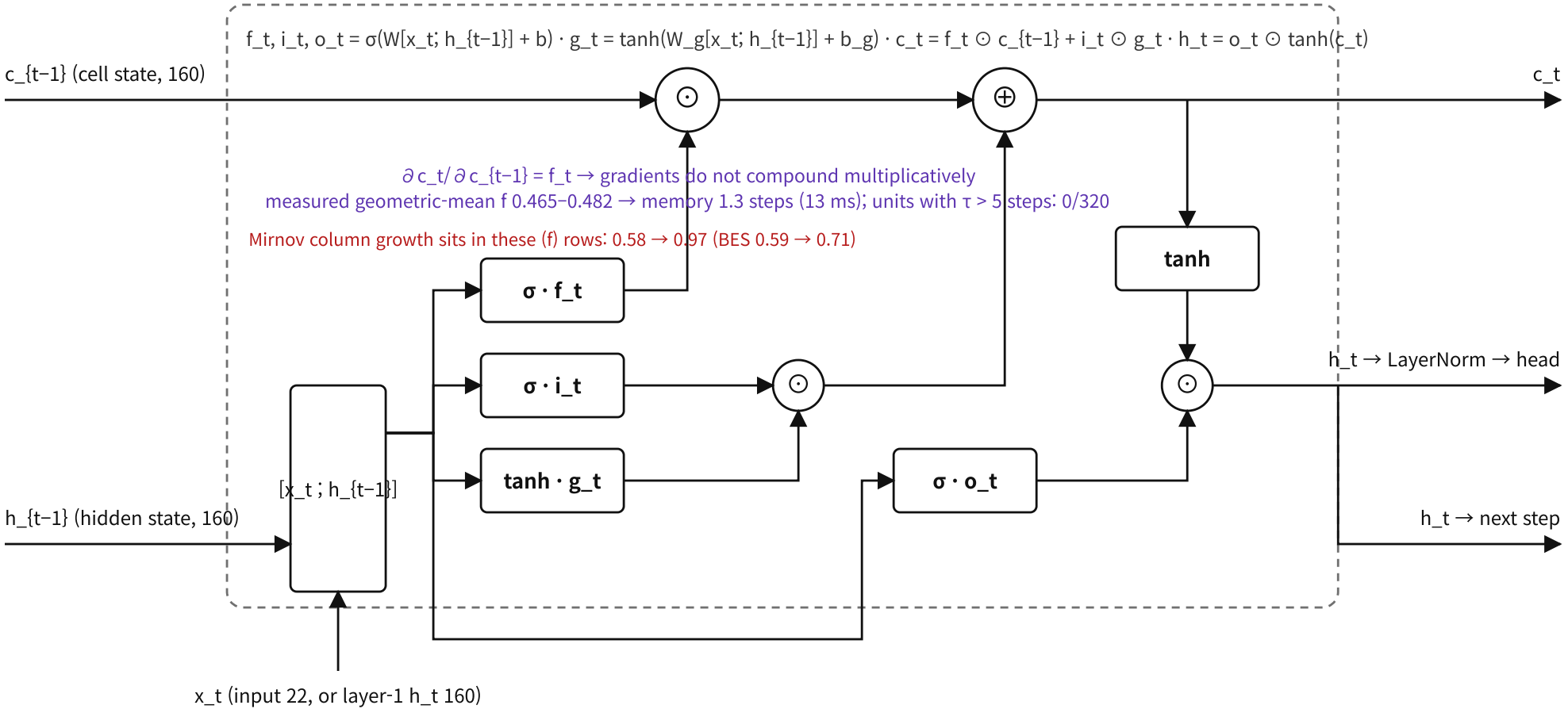

f_t, i_t, o_t = σ(W[x_t; h_{t−1}] + b) · g_t = tanh(W_g[x_t; h_{t−1}] + b_g) · c_t = f_t ⊙ c_{t−1} + i_t ⊙ g_t · h_t = o_t ⊙ tanh(c_t)
c_{t−1} (cell state, 160)
c_t
∂ c_t/ ∂ c_{t−1} = f_t → gradients do not compound multiplicatively
measured geometric-mean f 0.465–0.482 → memory 1.3 steps (13 ms); units with τ > 5 steps: 0/320
Mirnov column growth sits in these (f) rows: 0.58 → 0.97 (BES 0.59 → 0.71)
σ · f_t
σ · i_t
tanh · g_t
σ · o_t
tanh
[x_t ; h_{t−1}]
h_t → LayerNorm → head
h_t → next step
h_{t−1} (hidden state, 160)
x_t (input 22, or layer-1 h_t 160)